

Example 1. Find $f_y(1, 0)$ if:

$$f(x, y) = \sin \left[(1 - x + y)(2 - 2x + y) \cdots (10 - 10x + y) \right]$$

Solution. For a y partial we hold x constant at its value and take a y -derivative.

$$f_y(1, y) =$$

$$\text{And thus: } f_y(1, 0) =$$

Example 2. Let $f(x, y) = ye^{2xy}$ and find the following.

$$\frac{\partial f}{\partial x} = f_x(x, y) \text{ and } \frac{\partial f}{\partial y} = f_y(x, y)$$

Solution. This is alternative notation emphasizing the partial derivative as a:

For an x -partial we treat y as constant and then take an x -derivative.

$$\frac{\partial f}{\partial x} =$$

For a y -partial we treat x as constant and then take a y -derivative.

$$\frac{\partial f}{\partial y} =$$

Example 3. Let $f(x, y, z) = y^2 \sin(x) + e^z$ and find:

$$\frac{\partial f}{\partial x} = f_x(x, y, z) \text{ and } \frac{\partial f}{\partial y} = f_y(x, y, z) \text{ and } \frac{\partial f}{\partial z} = f_z(x, y, z)$$

Solution. For an x -partial we treat y and z as constant and take an x -derivative.

$$\frac{\partial f}{\partial x} =$$

For a y -partial we treat x and z as constant and take a y -derivative.

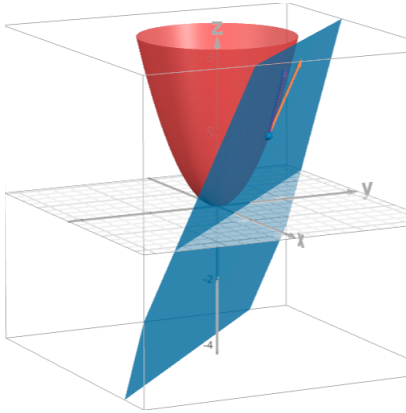
$$\frac{\partial f}{\partial y} =$$

For a z -partial we treat x and y as constant and take an z -derivative.

$$\frac{\partial f}{\partial z} =$$

A. **Tangent Planes.** Consider point **P** on a graph $z = f(x, y)$ with coordinates $x = a, y = b, z =$

The **tangent plane** at **P** rests flat against the graph at **P**.



The tangent vector at **P** contained in the xz cross-section is given by

$[x\text{-tangent}] =$

The tangent vector at **P** contained in the yz cross-section is given by

$[y\text{-tangent}] =$

Hence a normal to the tangent plane at **P** is given by the cross product

$\vec{n} = [x\text{-tangent}] \times [y\text{-tangent}] =$

The equation $\vec{n} \cdot (\vec{x} - \vec{p}) = 0$ of a plane can be applied.

We rearrange terms in the box below.

The **tangent plane** to $z = f(x, y)$ at point **P** with coordinates $x = a, y = b, z = f(a, b)$ is

or written more suggestively and less precisely

When we zoom in to the point of tangency, we see that the tangent plane:

On the graph, $z = f(x, y)$.

On the tangent plane, $z = L(x, y)$ where $L(x, y)$ is given below.

The **linear approximation** of $f(x, y)$ centered at $x = a$ and $y = b$ is

$L(x, y) =$

Example 4. Let $f(x, y) = ye^{2xy}$ and do the following.

(a) Find an equation for the tangent plane to $z = f(x, y)$ at the point $(0, 1, 1)$.

Solution. The point of interest has coordinates:

In an earlier example we calculated

$$f_x(x, y) = 2y^2e^{2xy}$$

$$f_y(x, y) = (1 + 2xy)e^{2xy}$$

Thus

$$f_x(0, 1) =$$

$$f_y(0, 1) =$$

The equation of the tangent plane reads $\Delta z = (\partial z / \partial x) \Delta x + (\partial z / \partial y) \Delta y$.

(b) Find a **rational number** approximating

$$0.9e^{0.18}$$

using a linear approximation of $f(x, y) = ye^{2xy}$ centered at:

A rational number is number that equals a fraction of integers. An integer is in the infinite list $0, \pm 1, \pm 2, \dots$

Solution. First we interpret the given value as an output of f .

$$0.9e^{0.18} = ye^{2xy}$$

for:

The linear approximation is found by solving for z in the tangent plane equation.

$$L(x, y) =$$

Next we utilize the approximation.

$$0.9e^{0.18} =$$